LOCAL QUASI-FINITENESS DESCENDS FROM A COMPOSITE MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the standard three-column reduction; the author writes that one applies the scheme statement to the top row, which is the whole argument.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Lifting a chain of morphisms of algebraic spaces to a chain of schemes. Spaces, Lemma spaces-lemma-lift-morphism-presentations, used at proof lines 5533-5541. Produces the three-column diagram.
- The scheme case: local quasi-finiteness descends from a composite. Morphisms, Lemma morphisms-lemma-permanence-quasi-finite, used at proof lines 5542-5544. Is the input applied to the top row.
- Local quasi-finiteness is etale local on source and target. used at proof lines 5532-5541. Justifies checking the property on the presentation.

Source reference. Original lemma 03WR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5122–5122. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02X1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3824–3824. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $X \to Y \to Z$ be morphisms of algebraic spaces over S. If $X \to Z$ is locally quasi-finite, then $X \to Y$ is locally quasi-finite.*

Proof. Choose a commutative diagram

$$\begin{array}{ccccc} U & \longrightarrow & V & \longrightarrow & W \\ \downarrow & & \downarrow & & \downarrow \\ X & \longrightarrow & Y & \longrightarrow & Z \end{array}$$

with vertical arrows étale and surjective. (See Spaces, Lemma 02X1.) Apply Morphisms, Lemma 03WR to the top row. □

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.